MILITARY SPECIFICATION

RADIOACTIVE GAMMA TEST SAMPLE, CS 137, M9

1. SCOPE

1.1 This specification covers a portable source of gamma radiation.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Steel, Strapping, Flat.
TT-V-121	-	Varnish, Spar, Water-Resisting.
PPP-F-291	-	Paperboard, Wrapping, Cushioning.
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-D-6055	-	Drums: Metal, Reuseable, Shipping and Storage.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

C124-12-19 - Marking Packing (Metal Shipping Container for RTS M9).
DL124-12-11 - Radioactive Gamma Test Sample, CS 137, M9.
D150-2-1 - Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM STANDARD

D999 - Vibration Test for Shipping Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL124-12-11 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components shall conform to the specifications and drawings listed on Drawing DL124-12-11 and subsidiary drawings.

3.1.2.1 Radiation source. The radiation source shall consist of 0.125 millicuries ± 10% of cesium 137, hermetically sealed in a capsule as specified on Drawing B124-12-14.

3.1.2.2 Radioactive leakage. The capsulated source shall show no leakage or transfer of radioactive material greater than 1×10^{-4} microcuries after rough handling for two hours at a force of at least one times gravity as specified in 4.2.4.1. The capsulated source shall be set aside for one week and retested only for removable activity.

3.1.2.3 Hermetic seal. The capsulated source shall show no evidence of air leakage when subjected to a heated glycerin bath for 60 seconds as specified in 4.2.4.2.

3.2 Assembly. The radioactive test sample shall be assembled in accordance with Drawing C124-12-11.

3.3 Radiation level. The test sample shall have a radiation level of 10 milliroentgens when tested as specified in 4.2.4.3.

3.4 External radiation. The gamma radiation shall not exceed 2 milliroentgens per hour (mr/hr) at any point on any surface of the assembled test sample when tested as specified in 4.2.4.4.

3.5 Radioactive contamination. The radioactive test sample shall show no contamination from radioactive material greater than 1×10^{-4} microcuries when tested as specified in 4.2.4.5.

3.6 Workmanship. The radioactive test sample shall be free from cracks, dents, chips in the body, nonfunctioning hardware, broken chain links and contamination such as grease or oil.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsibile for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the radioactive test samples produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions.

4.2.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive tests. The radioactive test samples shall be examined and tested in accordance with the classification of defects 4.2.3.3 and tested as specified in 4.2.4.

4.2.3.2 Examination for critical defects. Each item in the lot shall be inspected for critical characteristics listed in the classification of defects 4.2.3.3.

4.2.3.3 Classification of defects.

(a) Radioactive test sample Dwg C124-12-11.

Categories	Defects	Acceptance standards
Critical:		
1	Radiation level	4.2.4.3
2	External radiation	4.2.4.4
3	Marking missing, illegible, incorrect or incomplete	
4	Excessive radioactive contamination	4.2.4.5

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4.2.3.3 Classification of defects (continued).

(a) Radioactive test sample Dwg C124-12-11 (continued).

Categories	Defects
Major:	AQL 1.0 percent defective
101	Padlock missing or nonfunctioning
102	Chain length
103	Insert missing or loose fitting
104	Plug missing
105	Workmanship (see 3.6)

(b) CS 137 Source, Dwg B124-12-14.

Categories	Defects	Acceptance standards
Critical:		
1	Radioactive leakage	4.2.4.1
2	Hermetic seal	4.2.4.2
Major:	AQL 1.0 percent defective	
101	Threads incorrect	
102	0.31 dimension	

(c) Body, Dwg D124-12-12.

Categories	Defects
Major:	AQL 1.0 percent defective
101	4.500 dimension
102	1.505 dimension
103	2.250 dimension
104	1.255 dimension
105	3.00 dimension

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4.2.3.3 Classification of defects (continued).

(d) Preparation for delivery (section 5).

Categories	Defects
Critical:	
1	Marking of shipping container missing, illegible, incorrect, incomplete or not durable
Major:	AQL 1.0 percent defective
101	Shipping container not as specified
102	Cushioning of item in shipping container not as specified

4.2.4 Tests. Test shall be performed as follows:

4.2.4.1 Radioactive Leakage. Subject each capsulated radioactive source to a vibration test in accordance with ASTM Method D999 prior to assembly. However, for this test, replace the shipping container by a metal screw-cap container, measuring approximately 2 inches in diameter by 5 inches high. Place the capsule within the container without cushioning material and then place the container on the specified table. Vibrate the table for 2 hours at a speed that will yield output forces acting on the container equal to at least one times gravity. Remove the capsule from the container after vibration is complete and wipe all external surfaces with high wet-strength filter paper moistened with distilled water. Allow the filter paper to dry and then determine any contamination present by using a scaler equipped with scintillation or Geiger-Mueller tubes for counting. Counting equipment shall be capable of detecting 1×10^{-4} microcuries of radioactive material on the filter paper. Rubber gloves shall be worn for all wipe tests. Radioactivity shall not exceed 1×10^{-4} microcuries for each wipe test performed.

4.2.4.2 Hermetic seal. Immerse the capsulated radioactive source (Drawing B124-12-14) in glycerin or some other suitable liquid heated to 150° ± 5° C and allowed to remain in the bath for a minimum of 60 seconds. A steady stream of recurrent succession of bubbles from the sample indicates leakage.

4.2.4.3 Radiation level. Determine the radiation level of the test sample by inserting the external probe of a previously calibrated ionization meter into the well. The test sample shall produce a full scale reading of 10 milliroentgens on the low range scale of the instrument. The exposure time to obtain the full scale reading shall be 165 seconds minimum to 195 seconds maximum.

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4.2.4.4 External radiation. Check the external surface of the assembled test sample for radiation level using a suitable survey meter previously calibrated with a standard source. Observe for compliance with 3.4.

4.2.4.5 Radioactive contamination. Wipe throughly all accessible surfaces of the radioactive test sample, with a piece of high wet-strength filter paper moistened with distilled water. Allow the paper to dry and determine the radioactivity present using a scaler equipped with scintillation or G-M tubes for counting. Counting equipment shall be capable of detecting 1 x 10^{-4} microcuries of radioactive material on the filter paper.

5. PREPARATION FOR DELIVERY

5.1 Packing Level A. One test sample shall be packed in accordance with radioactive material and radiation limits specified in Department of Transportation Regulation (DOT) 49 CFR 171-179 in a type II, class S, size 6, drum of MIL-D-6055. The sides and bottom of the drum shall be cushioned by paperboard sleeve and fiberboard disc, respectively, of sufficient thickness to prevent movement of the sample in the drum. In addition, a tube shall be formed from a minimum of three layers of paperboard, and shall be fitted around the full height of the top projection, shown as item No 5, drawing C124-12-11. The inside diameter of the tube shall be 1.7 + 0.2 -0 inch, and the height of the tube shall be 1-1/4 ± 1/8 inch. The sleeve and tube shall be formed from paperboard conforming to type III, style I of PPP-P-291, with the flutes parallel to the vertical dimension of the drum. The disc shall be formed from fiberboard conforming to type CF, class domestic, variety SW, grade 175, of PPP-F-320.

5.2 Marking. In addition to any special marking required by the contract or order, drums shall be marked as shown on Edgewood Arsenal Drawing C124-12-19.

6. NOTES

6.1 Intended use. The radioactive test sample covered by this specification is intended for use as a source of gamma radiation.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU(EA)

Project Number 6665-A216

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